SHOPPING APP

Step 1: Creating the product model

Code:

```
class Product {
  final String id;
  final String name;
  final double price;
  final String category;
  final String description;

  Product({
    required this.id,
    required this.name,
    required this.price,
    required this.category,
    required this.description,
  });
}
```

Step 2: Writing the main.dart file

Code:

```
import 'package:flutter/material.dart';
import 'screens/home_screen.dart';
import 'screens/product_list_screen.dart';
import 'screens/product_details_screen.dart';
import 'screens/cart_screen.dart';
import 'screens/profile_screen.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
```

```
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Shopping Application',
      theme: ThemeData(
        primarySwatch: Colors.indigo,
      ),
      // Define initial route
      initialRoute: '/',
      // Define named route mapping
      routes: {
        '/': (context) => const HomeScreen(),
        '/products': (context) => const ProductListScreen(),
        '/details': (context) => const ProductDetailsScreen(),
        '/cart': (context) => const CartScreen(),
        '/profile': (context) => const ProfileScreen(),
      },
      debugShowCheckedModeBanner: false,
    );
  }
}
```

Step 3: Creating the Homescreen with a drawer

Code:

```
import 'package:flutter/material.dart';

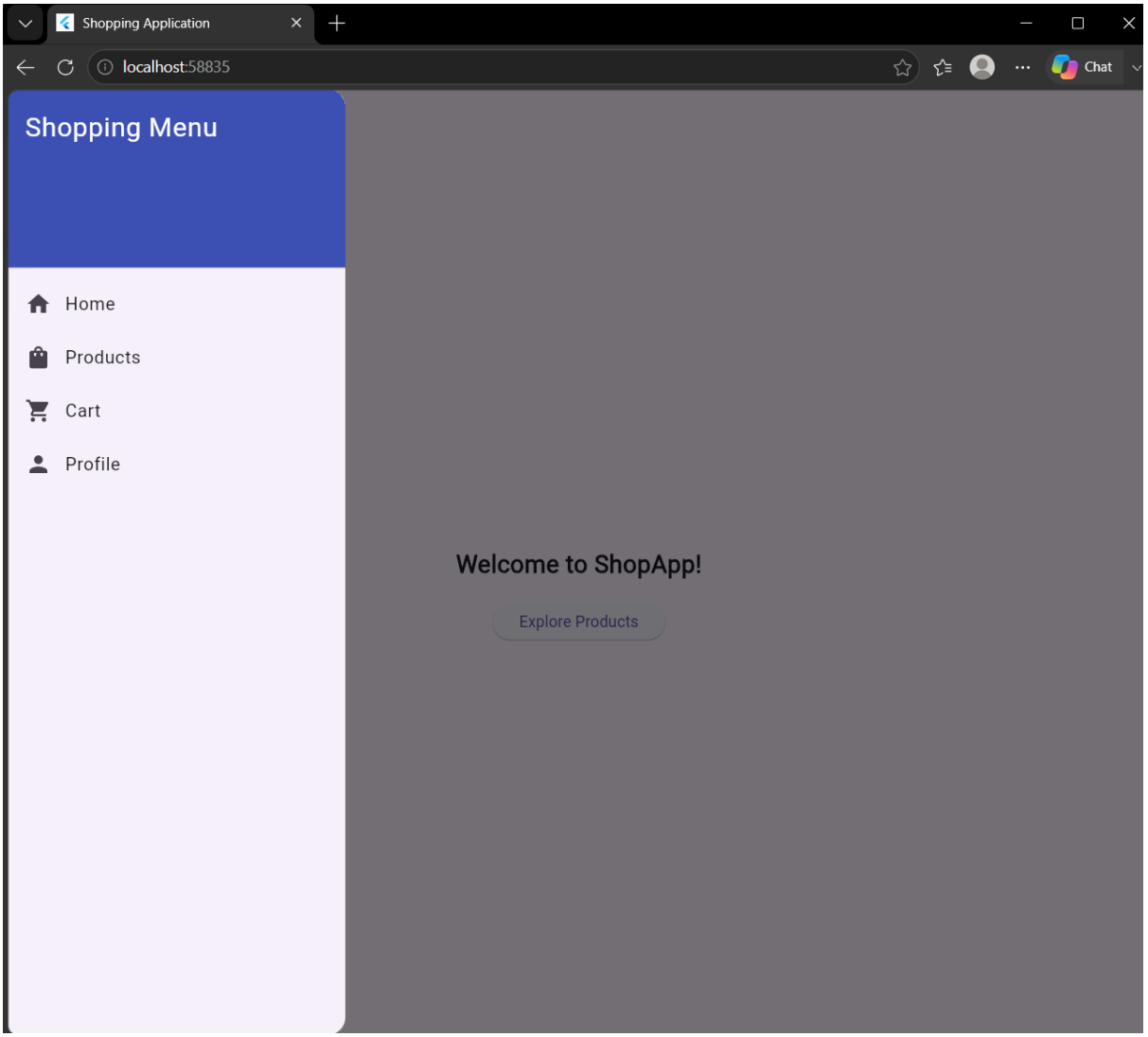
class HomeScreen extends StatelessWidget {
  const HomeScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Shopping App - Home'),
      ),
      drawer: Drawer(
        child: ListView(
          padding: EdgeInsets.zero,
          children: [
            const DrawerHeader(
              decoration: BoxDecoration(color: Colors.indigo),
              child: Text(
                'Shopping Menu',
                style: TextStyle(color: Colors.white, fontSize: 24),
              ),
```

```
      ),
      ListTile(
        leading: const Icon(Icons.home),
        title: const Text('Home'),
        onTap: () => Navigator.pop(context), // Close drawer
      ),
      ListTile(
        leading: const Icon(Icons.shopping_bag),
        title: const Text('Products'),
        onTap: () {
          Navigator.pop(context);
          Navigator.pushNamed(context, '/products');
        },
      ),
      ListTile(
        leading: const Icon(Icons.shopping_cart),
        title: const Text('Cart'),
        onTap: () {
          Navigator.pop(context);
          Navigator.pushNamed(context, '/cart');
        },
      ),
      ListTile(
        leading: const Icon(Icons.person),
        title: const Text('Profile'),
        onTap: () {
          Navigator.pop(context);
          Navigator.pushNamed(context, '/profile');
        },
      ),
    ],
  ),
),
body: Center(
  child: Column(
    mainAxisAlignment: MainAxisAlignment.center,
    children: [
      const Text(
        'Welcome to ShopApp!',
        style: TextStyle(fontSize: 22, fontWeight: FontWeight.bold),
      ),
      const SizedBox(height: 20),
      ElevatedButton(
        onPressed: () {
          // Navigate to Product List using named route
          Navigator.pushNamed(context, '/products');
        },
        child: const Text('Explore Products'),
```

```
            ),
          ],
        ),
      ),
    );
  }
}
```

Output:



Step 4: Creating the product list screen

Code:

```dart
import 'package:flutter/material.dart';
import '../models/products.dart';
```

```dart
class ProductListScreen extends StatelessWidget {
  ProductListScreen({super.key});

  // Dummy product list grouped by categories
  final List<Product> _allProducts = [
    Product(
      id: '1',
      name: 'Laptop',
      price: 45000.0,
      category: 'Electronics',
      description: 'High performance laptop for coding and tasks.',
    ),
    Product(
      id: '2',
      name: 'Smartphone',
      price: 20000.0,
      category: 'Electronics',
      description: 'Latest smartphone with great battery life.',
    ),
    Product(
      id: '3',
      name: 'T-Shirt',
      price: 799.0,
      category: 'Clothing',
      description: 'Comfortable cotton casual wear t-shirt.',
    ),
    Product(
      id: '4',
      name: 'Jeans',
      price: 1499.0,
      category: 'Clothing',
      description: 'Classic fit blue denim jeans.',
    ),
  ];

  @override
  Widget build(BuildContext context) {
    return DefaultTabController(
      length: 2,
      child: Scaffold(
        appBar: AppBar(
          title: const Text('Product Catalog'),
          bottom: const TabBar(
```

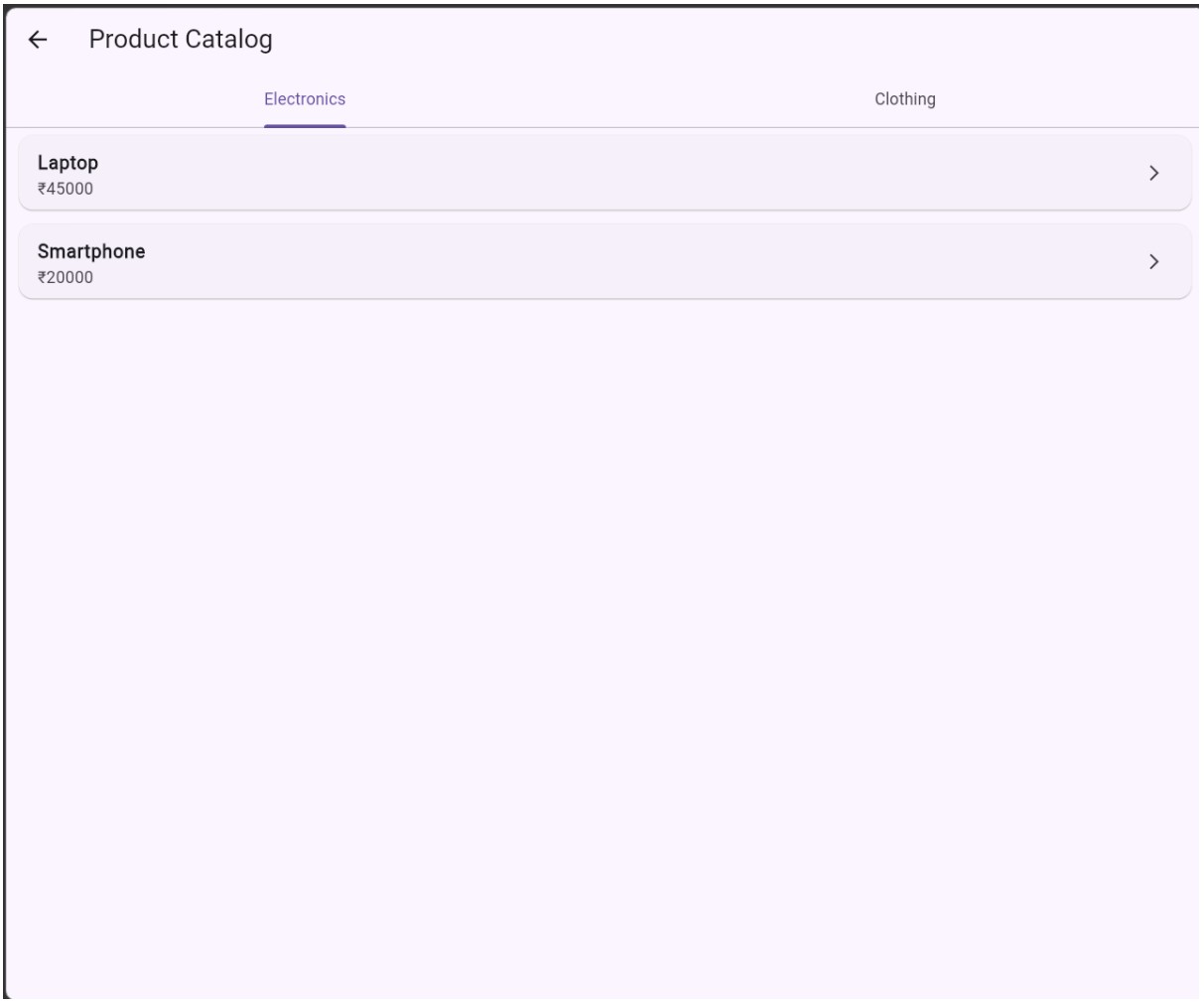
```
            tabs: [
              Tab(text: 'Electronics'),
              Tab(text: 'Clothing'),
            ],
          ),
        ),
        body: TabBarView(
          children: [
            // Electronics Category List
            _buildProductListView(context, 'Electronics'),
            // Clothing Category List
            _buildProductListView(context, 'Clothing'),
          ],
        ),
      ),
    );
  }

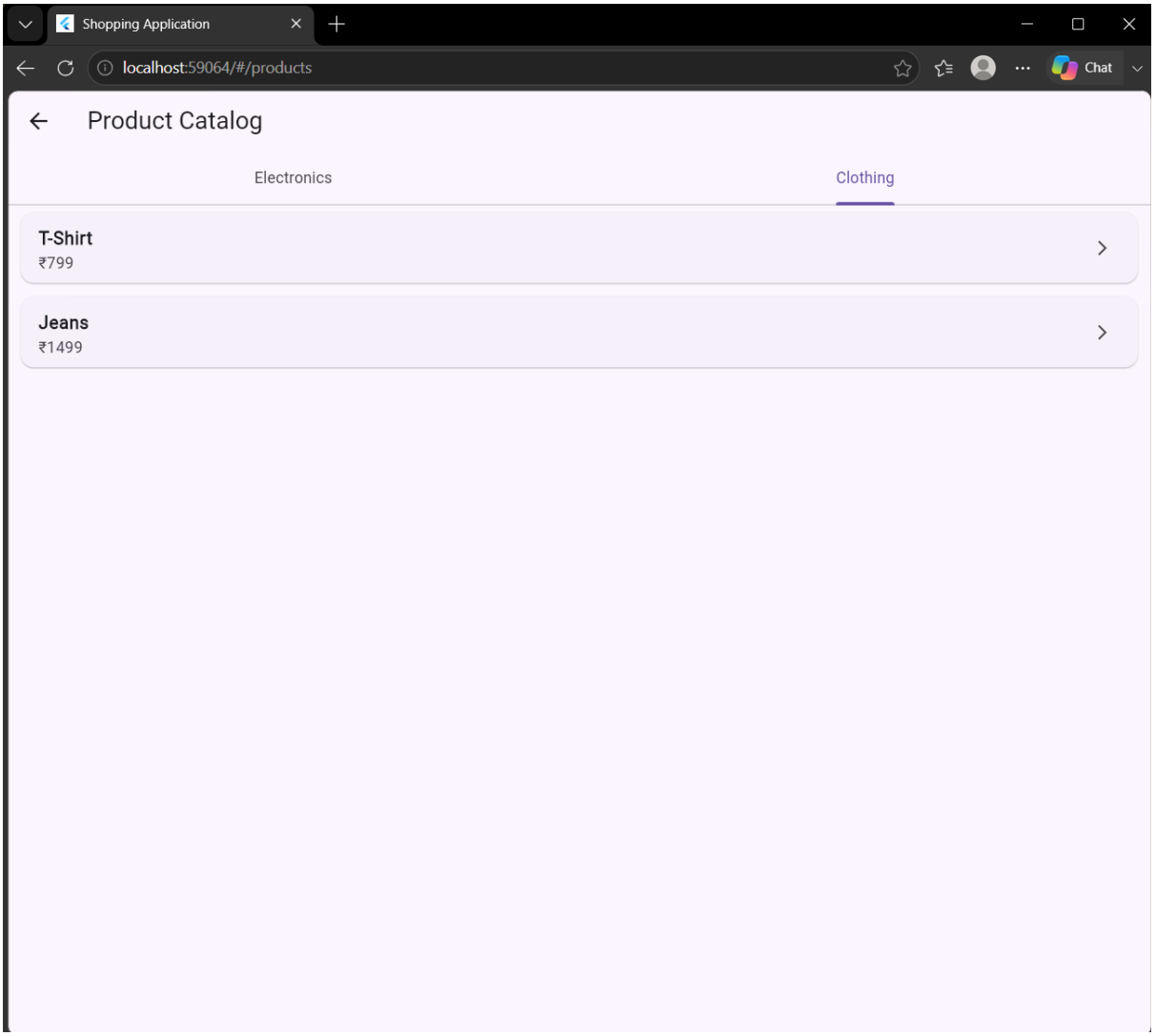
  Widget _buildProductListView(BuildContext context, String category) {
    final categoryProducts = _allProducts
        .where((p) => p.category == category)
        .toList();

    return ListView.builder(
      itemCount: categoryProducts.length,
      itemBuilder: (context, index) {
        final product = categoryProducts[index];
        return Card(
          margin: const EdgeInsets.symmetric(horizontal: 12, vertical:
6),
          child: ListTile(
            title: Text(
              product.name,
              style: const TextStyle(fontWeight: FontWeight.bold),
            ),
            subtitle: Text('₹${product.price}'),
            trailing: const Icon(Icons.arrow_forward_ios, size: 16),
            // Gesture / Tap interaction to open details and pass route
arguments
            onTap: () {
              Navigator.pushNamed(
                context,
                '/details',
```

```
                arguments: product, // Passing product object as route
argument
              );
            },
          ),
        );
      },
    );
  }
}
```

Output:





Step 5: Creating Product Details, Cart, and Profile Screen.

Code:

product_details_screen.dart

```
import 'package:flutter/material.dart';
import '../models/product.dart';

class ProductDetailsScreen extends StatelessWidget {
  const ProductDetailsScreen({super.key});

  @override
  Widget build(BuildContext context) {
    // Extract the product passed via route arguments
    final product = ModalRoute.of(context)!.settings.arguments as Product;

    return Scaffold(
      appBar: AppBar(
```

```
      title: Text(product.name),
    ),
    body: Padding(
      padding: const EdgeInsets.all(20.0),
      child: Column(
        crossAxisAlignment: CrossAxisAlignment.start,
        children: [
          Text(product.name, style: const TextStyle(fontSize: 24, fontWeight:
FontWeight.bold)),
          const SizedBox(height: 10),
          Text('Category: ${product.category}', style: const TextStyle(fontSize: 16, color:
Colors.grey)),
          const SizedBox(height: 10),
          Text('₹${product.price}', style: const TextStyle(fontSize: 22, color: Colors.indigo,
fontWeight: FontWeight.bold)),
          const SizedBox(height: 20),
          Text(product.description, style: const TextStyle(fontSize: 16)),
          const Spacer(),
          SizedBox(
            width: double.infinity,
            child: ElevatedButton.icon(
              icon: const Icon(Icons.shopping_cart),
              label: const Text('Add to Cart'),
              onPressed: () {
                // Navigate to Cart screen, passing the selected product along
                Navigator.pushNamed(
                  context,
                  '/cart',
                  arguments: product,
                );
              },
            ),
          ),
        ],
      ),
    ),
  );
 }
}
```

cart_screen.dart

```
import 'package:flutter/material.dart';
import '../models/product.dart';

class CartScreen extends StatelessWidget {
 const CartScreen({super.key});

 @override
```

```
  Widget build(BuildContext context) {
    // Retrieve product if passed from details screen
    final product = ModalRoute.of(context)!.settings.arguments as Product?;

    return Scaffold(
      appBar: AppBar(
        title: const Text('Shopping Cart'),
      ),
      body: product == null
          ? const Center(child: Text('Your cart is empty!'))
          : Padding(
              padding: const EdgeInsets.all(16.0),
              child: Column(
                children: [
                  Card(
                    child: ListTile(
                      title: Text(product.name, style: const TextStyle(fontWeight: FontWeight.bold)),
                      subtitle: Text('Category: ${product.category}'),
                      trailing: Text('₹${product.price}', style: const TextStyle(fontWeight:
FontWeight.bold, color: Colors.indigo)),
                    ),
                  ),
                  const Spacer(),
                  ElevatedButton(
                    onPressed: () {
                      ScaffoldMessenger.of(context).showSnackBar(
                        const SnackBar(content: Text('Order Placed Successfully!')),
                      );
                      Navigator.pushNamedAndRemoveUntil(context, '/', (route) => false);
                    },
                    child: const Text('Proceed to Checkout'),
                  ),
                ],
              ),
            ),
    );
  }
}
```

profile_screen.dart

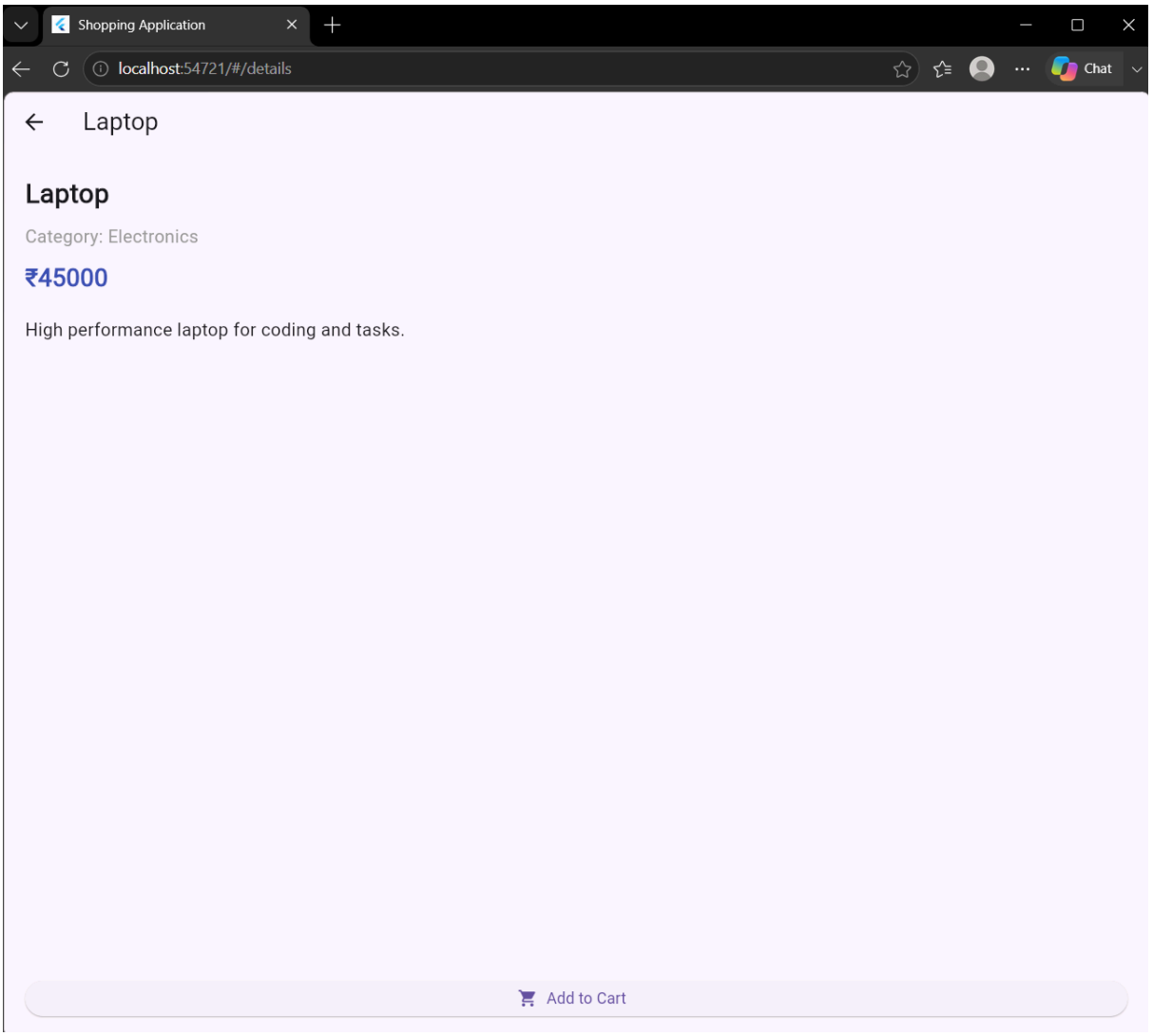
```
import 'package:flutter/material.dart';

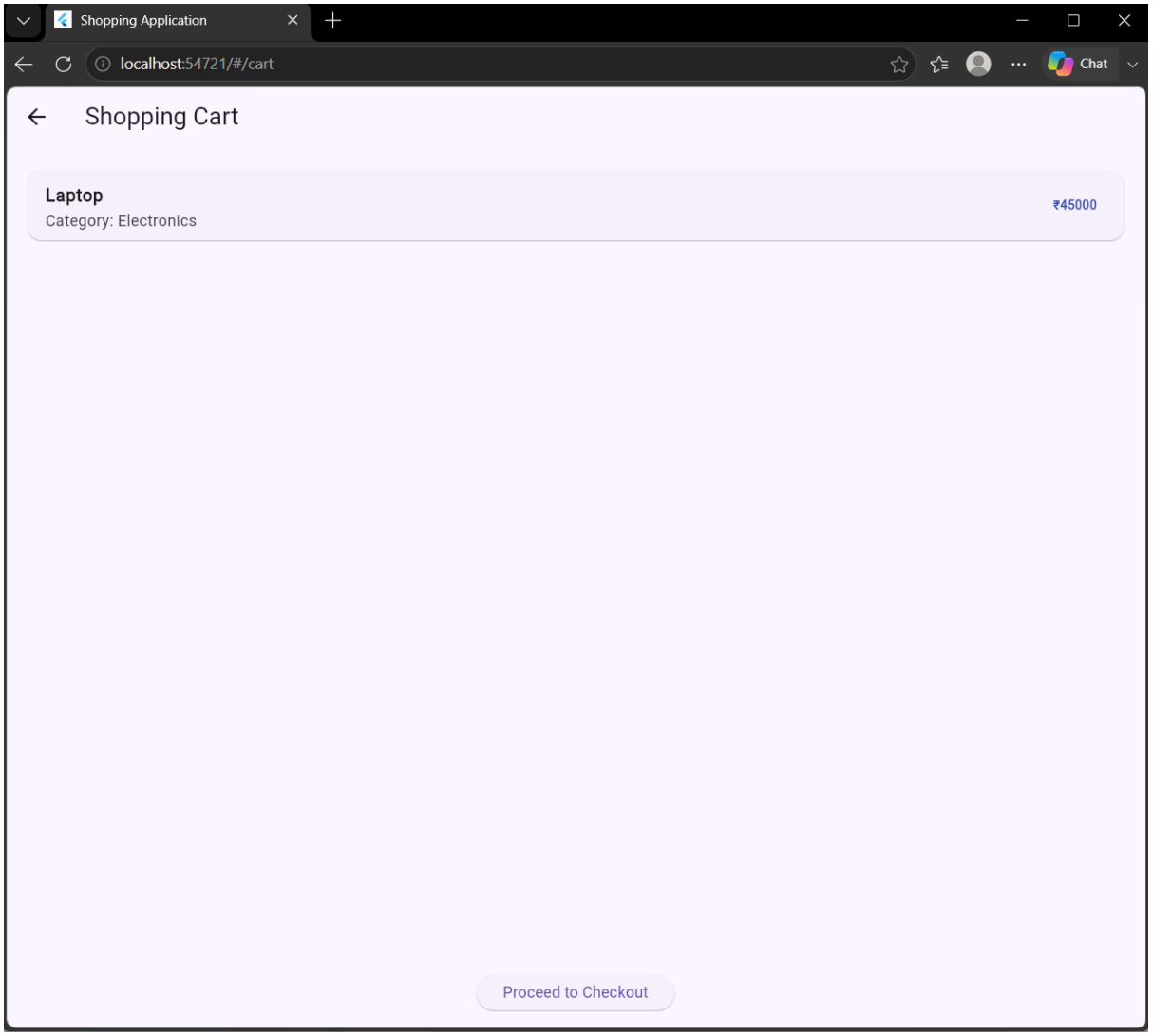
class ProfileScreen extends StatelessWidget {
  const ProfileScreen({super.key});

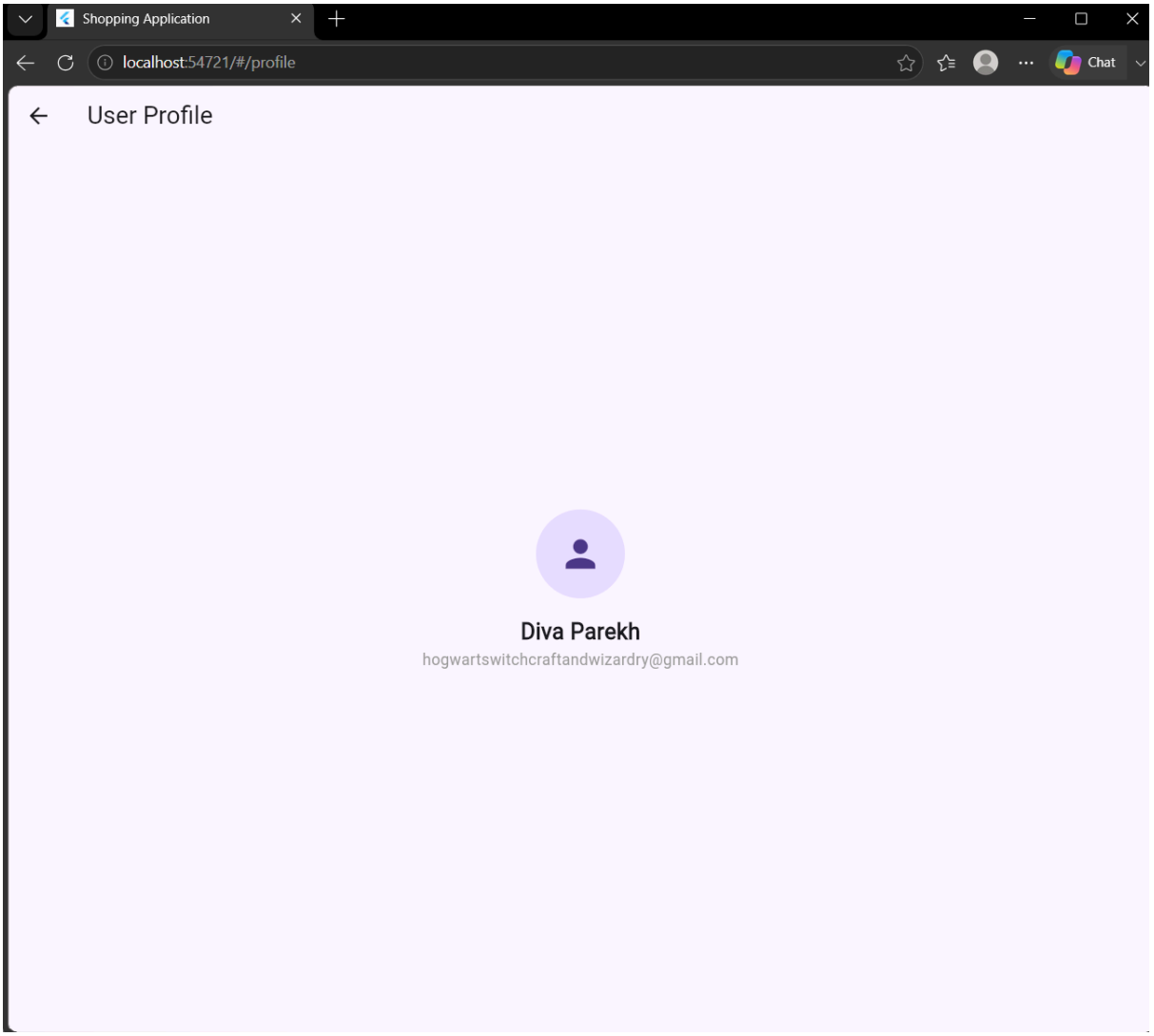
  @override
  Widget build(BuildContext context) {
    return Scaffold(
```

```
      appBar: AppBar(
        title: const Text('User Profile'),
      ),
      body: const Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            CircleAvatar(radius: 40, child: Icon(Icons.person, size: 40)),
            SizedBox(height: 16),
            Text('Diva Parekh', style: TextStyle(fontSize: 20, fontWeight: FontWeight.bold)),
            Text('student@marwadiuniversity.ac.in', style: TextStyle(color: Colors.grey)),
          ],
        ),
      ),
    );
  }
}
```

Output:


Shopping Application
localhost:54721/#/details
Chat
Laptop
Laptop
Category: Electronics
₹45000
High performance laptop for coding and tasks.
Add to Cart


Shopping Application
localhost:54721/#/cart
Chat
Shopping Cart
Laptop
Category: Electronics
₹45000
Proceed to Checkout


Shopping Application
localhost:54721/#/profile
Chat
User Profile
Diva Parekh
hogwartswitchcraftandwizardry@gmail.com